Zhaonan Qu

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Research Interests

My research interests are at the intersection of econometrics, operations research, and machine learning. I leverage novel connections between these topics to develop robust and scalable data-driven solutions in applications such as digital platforms, consumer preference, transportation, supply chains, energy networks, and finance.

Education

Ph.D in Economics (Econometrics), Stanford University **2024**
Minor: Operations Research
Dissertation: *Topics in Econometrics and Optimization*
Advisor: Guido Imbens (2021 Nobel Prize in Economics)
Committee: Alfred Galichon, Han Hong, Yinyu Ye, Johan Ugander

M.A. in Mathematics, Courant Institute, New York University **2017**

A.B. in Mathematics, Princeton University **2015**
Magna Cum Laude

Experience

Postdoctoral Research Scientist **2024–2026**
Data Science Institute, Columbia University
Mentors: Agostino Capponi, Simon Lee

Select Publications

On Sinkhorn's Algorithm and Choice Modeling
Operations Research (2026)
Qu, Zhaonan, Alfred Galichon, Wenzhi Gao, and Johan Ugander

Triply Robust Panel Estimators
Journal of Applied Econometrics (2026) [Python Package]
Athey, Susan, Guido Imbens, Zhaonan Qu, and Davide Viviano

Semiparametric Estimation of Treatment Effects in Observational Studies with Heterogeneous Partial Interference
Journal of Business & Economic Statistics (2026)
Qu, Zhaonan, Ruoxuan Xiong, Jizhou Liu, and Guido Imbens

Handling Sparse Non-negative Data in Business and Economics
EC '26: Proceedings of the 27th ACM Conference on Economics and Computation (2026)
Capponi, Agostino and Zhaonan Qu

Optimal Diagonal Preconditioning
Operations Research (2025)
Qu, Zhaonan, Wenzhi Gao, Oliver Hinder, Yinyu Ye, and Zhengyuan Zhou

Inferring Dynamic Networks from Marginals with Iterative Proportional Fitting
41st International Conference on Machine Learning (ICML) (2024)
Chang, Serina, Frederic Koehler, Zhaonan Qu, Jure Leskovec, and Johan Ugander

Federated Learning's Blessing: Fedavg has Linear Speedup
ICLR 2021 Workshop on Distributed and Private Machine Learning (DPML) (2021)
Qu, Zhaonan, Kaixiang Lin, Zhaojian Li, and Jiayu Zhou

Ensemble Methods for Causal Effects in Panel Data Settings
American Economic Association Papers and Proceedings 109: 65-70 (2019)
Athey, Susan, Mohsen Bayati, Guido Imbens, and Zhaonan Qu

Working Papers

Learning Heterogeneous Contextual Effects in Ordered Choice Sets: An Application to Online Marketplaces
Working Paper (2026)
Qu, Zhaonan and Patrick Ding

Sharp Non-Asymptotic Local Convergence and Acceleration of Sinkhorn's Algorithm
Under Review (2026)
Gao, Wenzhi, Zhaonan Qu, Madeleine Udell, and Yinyu Ye

Structured Lasso for Convex Nonparametric Least Squares: An Application to Swedish Electricity Distribution Networks
Major Revision at *European Journal of Operational Research* (2025)
Liao, Zhiqiang and Zhaonan Qu

Distributionally Robust Instrumental Variables Estimation
Under Review (2024)
Qu, Zhaonan and Yongchan Kwon

Ongoing Projects

Adaptive Learning of Consumer Preferences: A Graph Connectivity Optimization Approach
Gao, Wenzhi and Zhaonan Qu

Estimating the Value of Sponsored Ads in Search Results
Zhaonan Qu

Teaching Experience

MGMT 416 AI for Business Decisions (Instructor), NJIT Fall 2025
MS&E 311 Optimization (TA), Stanford Winter 2021
ECON 292 Quantitative Methods for Empirical Research (TA), Stanford Autumn 2020

Invited Presentations

EC '26: 27th ACM Conference on Economics and Computation	July 2026
Annual Conference of the International Association for Applied Econometrics (IAAE)	June 2026
INFORMS Optimization Society Conference (IOS)	March 2026
International Conference on Information Technology and Quantitative Management (ITQM)	August 2025
International Conference on Continuous Optimization (ICCOPT)	July 2025
INFORMS Service Science Conference, Said Business School, Oxford	July 2025
Annual Conference of the International Association for Applied Econometrics (IAAE)	June 2025
INFORMS Annual Meeting Session on Financial Analytics and Technology	October 2024
41st International Conference on Machine Learning (ICML)	July 2024
2024 American Causal Inference Conference Session on Instrumental Variables	May 2024
2024 Banff Research Center Workshop on Optimal Transport and Distributional Robustness	March 2024
INFORMS Annual Meeting Session on Econometric, Big Data Methods and Applications to Finance	October 2023
INFORMS Annual Meeting Poster Session on Operations Research and Optimization Methodologies	October 2023
2023 Stanford Data Science Conference Poster Session	May 2023
2022 North American Summer Meeting of the Econometric Society	June 2022
2022 American Causal Inference Conference Poster Session	May 2022
2022 California Econometrics Conference Poster Session	May 2022
INFORMS Annual Meeting Session on Stochastic Optimization (virtual)	November 2020

Research Affiliations, Honors, and Awards

Arnold Ventures Research Fellowship in Jail Data Initiative Social Science Research Council	2024–2026
Postdoctoral Research Assistant Professor Johan Ugander's Lab	2023–2024
Stanford Interdisciplinary Graduate Fellowship Stanford University	2019–2022
Provost's Global Research Initiative Fellowship, New York University	2016

Magna Cum Laude in Mathematics and Phi Beta Kappa, Princeton University 2015
7th Place, Rotman International Trading Competition, University of Toronto 2014
Smith-Newton Scholar, Princeton Environmental Institute 2013

Peer Review Services

Journal of Econometrics, Statistics in Medicine, Mathematical Programming Computation, SIAM Journal on Matrix Analysis and Applications, Decision Sciences Journal, Journal of Scientific Computing, Journal of Artificial Intelligence Research, Journal of Computational and Graphical Statistics, SIAM Journal on Financial Mathematics

Programming Languages

Python, MATLAB, SQL, R

Internships

Microsoft Research New England, Research Data Science Intern
Robustness and Causal Inference *2021*

Uber, Data Science Intern
Personalized Tipping Suggestions based on Trip Quality *2020*

Cruise, Data Scientist Intern
Learning Causal Bayesian Networks through Knockoffs *2019*